

Physical Chemistry (I)

Lecture 07

Thermochemistry



In the Last Class

- The first law of thermodynamics
 - Heat
 - Differential form of internal energy
- Enthalpy
 - State and path functions
 - Total differentials
 - Response functions



Last class

Heat

At constant volume,

$$\delta q_V \propto dT$$

$$\delta q_V = C_V dT$$

At constant pressure,

$$\delta q_p \propto dT$$

$$\delta q_p = C_p dT$$



Last class

Calculation of Heat (Const V)

$$\delta q_V = C_V dT$$

- Total heat for a change from **state i** to **state f**

$$q_V = \int_{\text{state } i}^{\text{state } f} \delta q_V = \int_{\text{state } i}^{\text{state } f} (C_V dT) = \int_{T_i}^{T_f} C_V dT$$

First Law Revisited

$$dU = \delta q + \delta w$$

For $\delta w_{\text{add}} = 0$,

$$dU = C dT - p_{\text{ex}} dV$$

At constant volume, $dV = 0$ and $C = C_V$

$$dU = C_V dT$$

If C_V is constant,

$$\Delta U = C_V \Delta T$$

State and Path Functions

- **State function:** a property that depends only on the current state of a system and is independent of its history
e.g. internal energy, pressure, volume, temperature
- **Path function:** a property with a value that depends on the path between two states
e.g. heat, work

Related Mathematics

- **Exact differential:** an infinitesimal quantity that, when integrated, gives a result that is independent of the path between the initial and final states.

$$\Delta U = \int_{\text{state } i}^{\text{state } f} dU$$

- **Inexact differential:** an infinitesimal quantity that, when integrated, gives a result that depends on the path between the initial and final states.

$$q = \int_{\text{state } i, \text{ path}}^{\text{state } f} \delta q$$

Use of Differentials

$$dU = \underbrace{\left(\frac{\partial U}{\partial V}\right)_T}_{\pi_T} dV + \underbrace{\left(\frac{\partial U}{\partial T}\right)_V}_{C_V} dT$$

$$dU = \pi_T dV + C_V dT$$

At constant pressure,

$$\left(\frac{\partial U}{\partial T}\right)_p = \pi_T \left(\frac{\partial V}{\partial T}\right)_p + C_V$$

Enthalpy H

$$H = U + pV$$

- Energy required to create a system (U) and to make a room for the system (pV)
- Meaning: $dH = \delta q_p$ or $\Delta H = q_p$

$$C_p = \left(\frac{\partial H}{\partial T} \right)_p$$

Differential Form of H

For $H = H(p, T)$,

$$dH = \left(\frac{\partial H}{\partial p} \right)_T dp + \left(\frac{\partial H}{\partial T} \right)_p dT$$

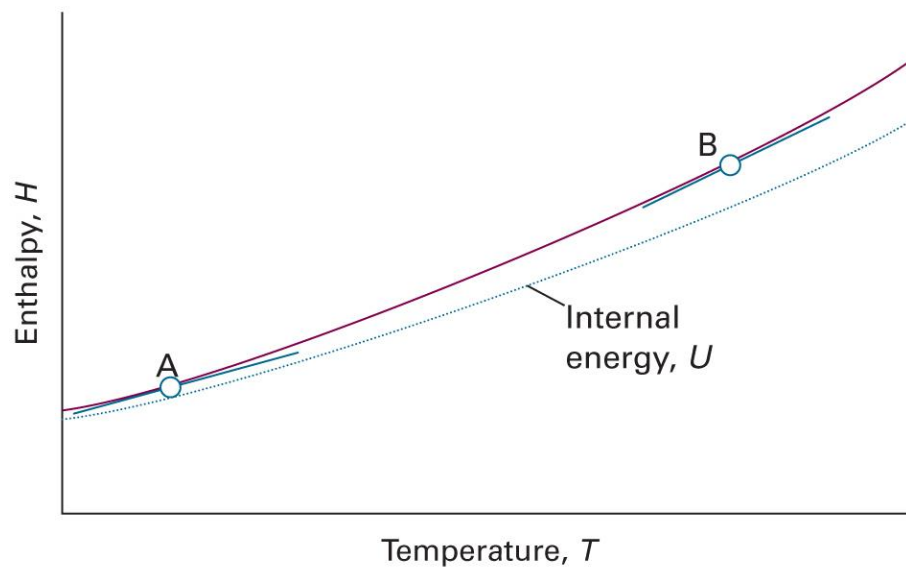
$$dH = -C_p \mu dp + C_p dT$$

(Compare: $dU = \pi_T dV + C_V dT$)

Last class

C_V and C_p

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V, \quad C_p = \left(\frac{\partial H}{\partial T} \right)_p$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 2B.3)

Adiabatic Changes

$$\delta q = 0$$

$$dU = \delta q + \delta w = \delta w$$

“Adiabatic expansion” $\rightarrow U$ drop $\rightarrow T$ drop

Work for Perfect Gas

$$dU = \delta w$$

For a perfect gas, $dU = C_V dT$

$$C_V dT = \delta w$$

Integration leads to

$$w = \int_{\text{state } i}^{\text{state } f} \delta w = \int_{\text{state } i}^{\text{state } f} C_V dT = \int_{T_i}^{T_f} C_V dT$$

If C_V is constant,

$$w = C_V (T_f - T_i)$$

Perfect Gas, Reversible Change

$$dU = \delta w$$

- Perfect gas: $dU = C_V dT$
- Reversible change: $\delta w = -pdV$

$$C_V dT = -pdV$$

Since $p = nRT/V$,

$$\frac{C_V}{T} dT = -\frac{nR}{V} dV$$

Integration

$$\int_{\text{state } i}^{\text{state } f} \frac{C_V}{T} dT = - \int_{\text{state } i}^{\text{state } f} \frac{nR}{V} dV$$

$$\int_{T_i}^{T_f} \frac{C_V}{T} dT = - \int_{V_i}^{V_f} \frac{nR}{V} dV$$

If C_V is constant,

$$C_V \int_{T_i}^{T_f} \frac{1}{T} dT = -nR \int_{V_i}^{V_f} \frac{1}{V} dV$$

$$C_V (\ln T_f - \ln T_i) = -nR (\ln V_f - \ln V_i)$$

Simplification

$$C_V(\ln T_f - \ln T_i) = -nR(\ln V_f - \ln V_i)$$

$$\frac{C_V}{nR} \ln \frac{T_f}{T_i} = -\ln \frac{V_f}{V_i} = \ln \frac{V_i}{V_f}$$

Let $c = C_V/nR$;

$$\frac{C_V}{nR} \ln \frac{T_f}{T_i} = c \ln \frac{T_f}{T_i} = \ln \left(\frac{T_f}{T_i} \right)^c$$

$$\left(\frac{T_f}{T_i} \right)^c = \frac{V_i}{V_f}$$

Volume-Temperature

For $c = C_V/nR$,

$$\left(\frac{T_f}{T_i}\right)^c = \frac{V_i}{V_f}$$

Therefore,

$$V_i T_i^c = V_f T_f^c$$

or,

$$VT^c = \text{constant}$$

From T to p

$$V_i T_i^c = V_f T_f^c$$

For a perfect gas, $T = pV/nR$

$$V_i \left(\frac{p_i V_i}{nR} \right)^c = V_f \left(\frac{p_f V_f}{nR} \right)^c$$

$$p_i^c V_i^{c+1} = p_f^c V_f^{c+1}$$

$$p_i V_i^{(c+1)/c} = p_f V_f^{(c+1)/c}$$

Volume-Pressure

For $c = C_V/nR$,

$$p_i V_i^{(c+1)/c} = p_f V_f^{(c+1)/c}$$

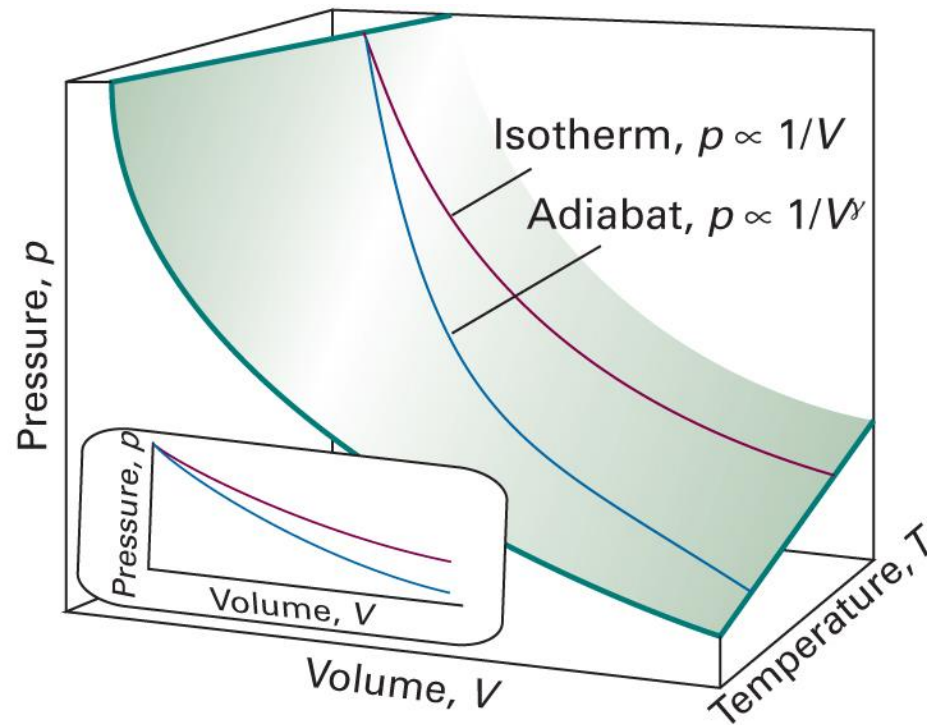
Let $\gamma = (c + 1)/c = 1 + 1/c$;

$$p_i V_i^\gamma = p_f V_f^\gamma$$

or,

$$pV^\gamma = \text{constant}$$

Isotherm and Adiabatic



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 2E.2)

Monatomic Perfect Gas

$$VT^c = \text{constant}$$

$$pV^\gamma = \text{constant}$$

$$c = \frac{C_V}{nR} = \frac{(3/2)nR}{nR} = \frac{3}{2}$$

$$\gamma = 1 + \frac{1}{c} = 1 + \frac{2}{3} = \frac{5}{3}$$

Summary for Chapter 3 (2)

Thermodynamics	Mathematics
State function	Exact differential
Path function	Inexact differential

Differential form of U

$$dU = \left(\frac{\partial U}{\partial V}\right)_T dV + \left(\frac{\partial U}{\partial T}\right)_V dT$$

Enthalpy

$$H = U + pV$$

$$dH = \delta q_p$$

Differential form of H

$$dH = \left(\frac{\partial H}{\partial p}\right)_T dp + \left(\frac{\partial H}{\partial T}\right)_p dT$$

$$C_V = (\partial U / \partial T)_V$$

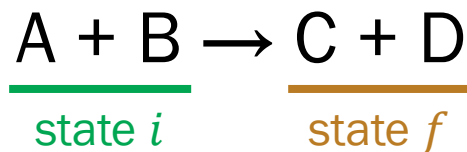
C_V and C_p

- Perfect gas
- Adiabatic change: $\delta q = 0$

$$C_p = (\partial H / \partial T)_p$$

Thermochemistry

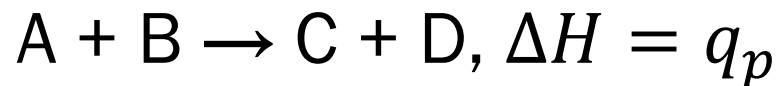
“The study of the energy transferred as heat during the course of chemical reactions”



$$q_V = \Delta U$$

$$q_p = \Delta H$$

Endothermic and Exothermic



$\Delta H > 0$: endothermic reaction

$\Delta H < 0$: exothermic reaction

Standard Enthalpy, $H^\ominus$

Enthalpy H of a substance in its **standard state**

- **Standard state** of a substance: its pure form at 1 bar
 - Defined for a specific temperature
 - If temperature is not specified, use the conventional temperature of 298.15 K
- Standard enthalpy change $\Delta H^\ominus = H_f^\ominus - H_i^\ominus$

Example: Vaporization of Water



$$\Delta H^\ominus(373 \text{ K}) = 40.66 \text{ kJ mol}^{-1}$$

pure liquid water $\rightarrow$ pure gas water

at 1 bar and 373 K

H and ΔH

$$\Delta H = H(\text{product}) - H(\text{reactant})$$

$$H = U + pV$$

One can determine absolute H for a perfect gas

But what about real substances?

(liquids, solids, solutions, ...)

Instead, we focus on the “change,” ΔH , in chemistry

For Typical Changes

- The standard enthalpy change is called the **standard enthalpy of [transition]**
- Physical change
 - Vaporization $\Delta_{\text{vap}}H^\ominus$: standard enthalpy of vaporization
 - Fusion $\Delta_{\text{fus}}H^\ominus$: standard enthalpy of fusion
- Chemical change
 - Reaction $\Delta_{\text{r}}H^\ominus$: standard enthalpy of reaction
 - Combustion $\Delta_{\text{c}}H^\ominus$: standard enthalpy of combustion

Several Types of Enthalpies

Transition	Process	Symbol*
Transition	Phase $\alpha \rightarrow$ phase β	$\Delta_{\text{trs}}H$
Fusion	$s \rightarrow l$	$\Delta_{\text{fus}}H$
Vaporization	$l \rightarrow g$	$\Delta_{\text{vap}}H$
Sublimation	$s \rightarrow g$	$\Delta_{\text{sub}}H$
Mixing	Pure $\rightarrow$ mixture	$\Delta_{\text{mix}}H$
Solution	Solute $\rightarrow$ solution	$\Delta_{\text{sol}}H$
Hydration	$X^+(g) \rightarrow X^+(aq)$	$\Delta_{\text{hyd}}H$
Atomization	Species(s, l, g) $\rightarrow$ atoms(g)	$\Delta_{\text{at}}H$
Ionization	$X(g) \rightarrow X^+(g) + e^-(g)$	$\Delta_{\text{ion}}H$
Electron gain	$X(g) + e^-(g) \rightarrow X^-(g)$	$\Delta_{\text{eg}}H$
Reaction	Reactants $\rightarrow$ products	Δ_rH
Combustion	Compound(s, l, g) + $O_2(g) \rightarrow CO_2(g) + H_2O(l, g)$	Δ_cH
Formation	Elements $\rightarrow$ compound	Δ_fH
Activation	Reactants $\rightarrow$ activated complex	$\Delta^\ddagger H$

(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Table 2C.2)



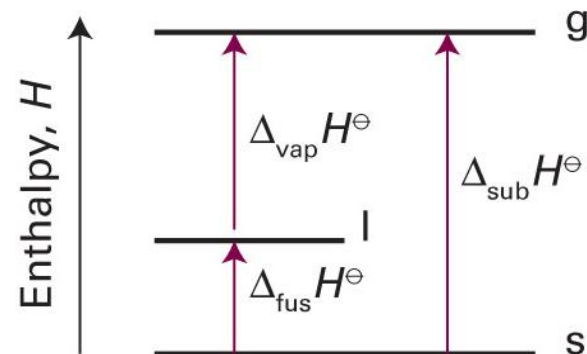
H as State Function

ΔH is always same if the initial and final states are same

Path 1: $\text{H}_2\text{O}(\text{s}) \rightarrow \text{H}_2\text{O}(\text{g})$, $\Delta_{\text{sub}}H^\ominus$

Path 2: (1) $\text{H}_2\text{O}(\text{s}) \rightarrow \text{H}_2\text{O}(\text{l})$, $\Delta_{\text{fus}}H^\ominus$

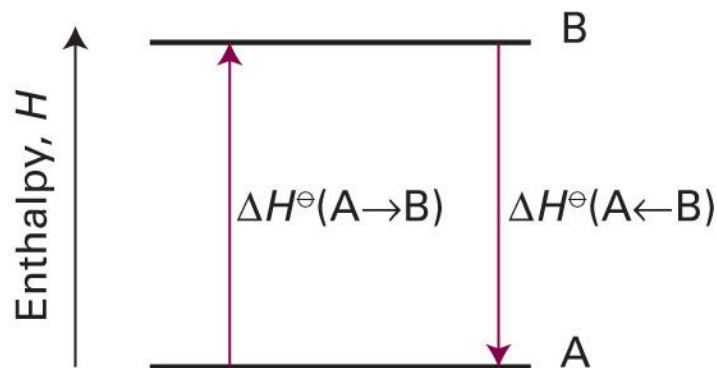
(2) $\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{O}(\text{g})$, $\Delta_{\text{vap}}H^\ominus$



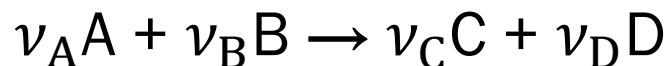
$$\Delta_{\text{sub}}H^\ominus = \Delta_{\text{fus}}H^\ominus + \Delta_{\text{vap}}H^\ominus$$

H as State Function

$$\Delta H^\ominus(\text{forward}) = -\Delta H^\ominus(\text{reverse})$$



Stoichiometry: Mole Method

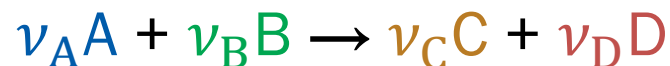


means

“ ν_A moles of A and ν_B moles of B react
to produce ν_C moles of C and ν_D moles of D.”

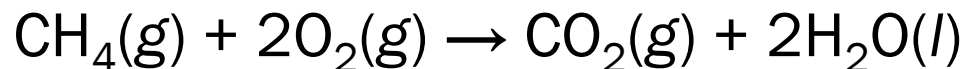
(ν_i : stoichiometric coefficients)

Standard Reaction Enthalpy



- Standard reaction enthalpy $\Delta_r H^\ominus$: standard enthalpy change $\Delta H^\ominus$ for the reaction per ...
 - ν_A moles of consumed A
 - ν_B moles of consumed B
 - ν_C moles of produced C
 - ν_D moles of produced D

Example



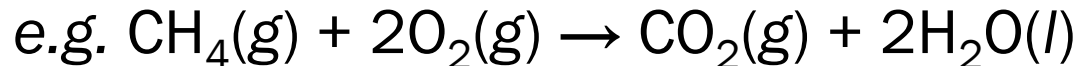
$$\Delta_{\text{r}}H^{\ominus} = -890 \text{ kJ mol}^{-1}$$

The enthalpy change is -890 kJ at 1 bar and 298.15 K per

- 1 mol of $\text{CH}_4(\text{g})$ in its pure form
- 2 mol of $\text{O}_2(\text{g})$ in its pure form
- 1 mol of $\text{CO}_2(\text{g})$ in its pure form
- 2 mol of $\text{H}_2\text{O}(\text{l})$ in its pure form

Formal Definition

$$\Delta_r H^\ominus = \sum_{\text{products } i} \nu(i) H_m^\ominus(i) - \sum_{\text{reactants } j} \nu(j) H_m^\ominus(j)$$



$$\begin{aligned} \Delta_r H^\ominus &= \nu(\text{CO}_2) H_m^\ominus(\text{CO}_2, \text{g}) + \nu(\text{H}_2\text{O}) H_m^\ominus(\text{H}_2\text{O}, \text{l}) \\ &\quad - \nu(\text{CH}_4) H_m^\ominus(\text{CH}_4, \text{g}) - \nu(\text{O}_2) H_m^\ominus(\text{O}_2, \text{g}) \\ &= H_m^\ominus(\text{CO}_2, \text{g}) + 2H_m^\ominus(\text{H}_2\text{O}, \text{l}) - H_m^\ominus(\text{CH}_4, \text{g}) - 2H_m^\ominus(\text{O}_2, \text{g}) \end{aligned}$$

Problem in $\Delta_r H^\ominus$

$$\Delta_r H^\ominus = \sum_{\text{products } i} \nu(i) H_m^\ominus(i) - \sum_{\text{reactants } j} \nu(j) H_m^\ominus(j)$$

Can we determine absolute $H_m^\ominus(i)$?

Hess's Law

“The standard reaction enthalpy is the sum of the values for the individual reactions into which the overall reaction may be divided.”

This holds because H is a **state function**

Example 2C.1

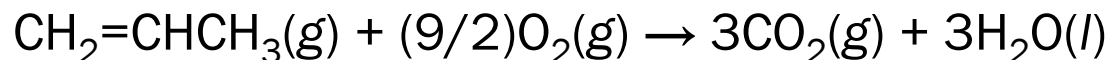
The standard reaction enthalpy for the hydrogenation of propene, $\text{CH}_2=\text{CHCH}_3(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{CH}_3\text{CH}_2\text{CH}_3(\text{g})$, is -124 kJ mol^{-1} .

The standard reaction enthalpy for the combustion of propane, $\text{CH}_3\text{CH}_2\text{CH}_3(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{l})$, is $-2220 \text{ kJ mol}^{-1}$.

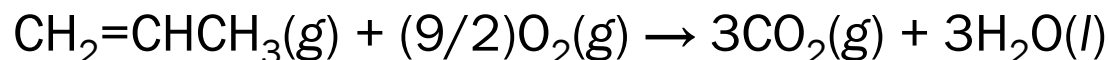
The standard reaction enthalpy for the formation of water, $\text{H}_2(\text{g}) + (1/2)\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$, is -286 kJ mol^{-1} .

Calculate the standard enthalpy of combustion of propene.

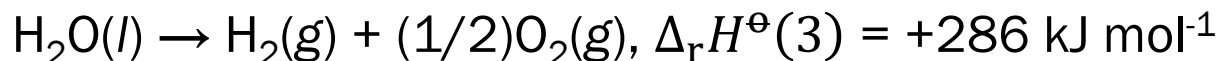
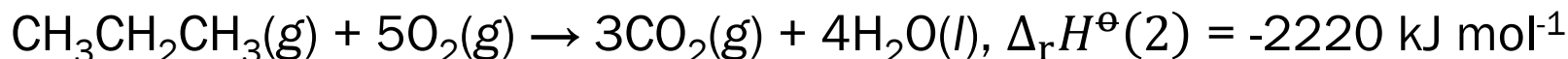
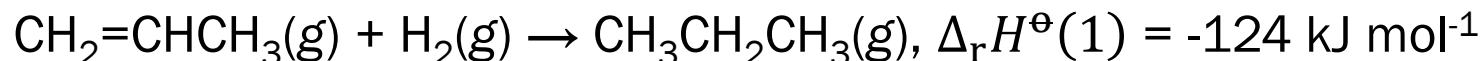
The target reaction is



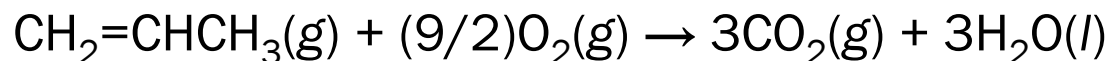
Example 2C.1



Arranging the given reactions, we can obtain



Summing up, the whole reaction is



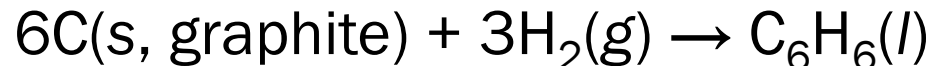
and

$$\Delta_{\text{r}}H^\ominus = \Delta_{\text{r}}H^\ominus(1) + \Delta_{\text{r}}H^\ominus(2) + \Delta_{\text{r}}H^\ominus(3) = -2058 \text{ kJ mol}^{-1}$$

Standard Enthalpy of Formation

- Standard enthalpy of formation $\Delta_f H^\ominus$ of a substance: $\Delta_r H^\ominus$ for the formation of the compound from its elements in their reference states
- Reference state of an element: its most stable state at the specified temperature and 1 bar

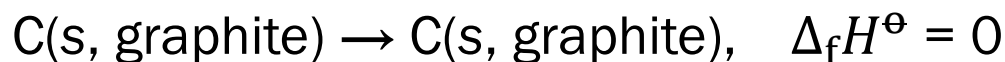
e.g. $\Delta_f H^\ominus$ of liquid benzene at 298 K: +49.0 kJ mol⁻¹



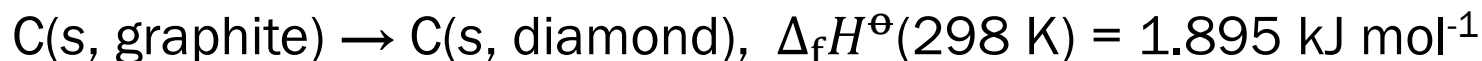
$\Delta_f H^\ominus$ of Elements

- For an element ...

- in its reference state, $\Delta_f H^\ominus = 0$ (for all T)



- in the other state, $\Delta_f H^\ominus > 0$



- Reference state of phosphorus (P): white phosphorus

- not the most stable, but the most reproducible



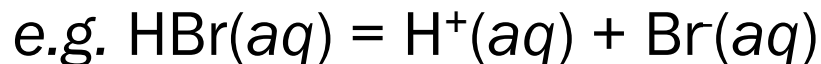
$\Delta_f H^\ominus$ of Ions

- $\Delta_f H^\ominus$ of ions: e.g. $\text{Na(s)} \rightarrow \text{Na}^+(\text{aq}) + \text{e}^-$

Impossible to isolate $\text{Na}^+(\text{aq})$

- Define $\Delta_f H^\ominus$ for one ion, and use it to determine others

$$\Delta_f H^\ominus(\text{H}^+, \text{aq}) = 0$$



$$\Delta_f H^\ominus(\text{HBr}, \text{aq}) = \Delta_f H^\ominus(\text{H}^+, \text{aq}) + \Delta_f H^\ominus(\text{Br}, \text{aq})$$

↑
measurable

= 0

Some Values of $\Delta_f H^\ominus$

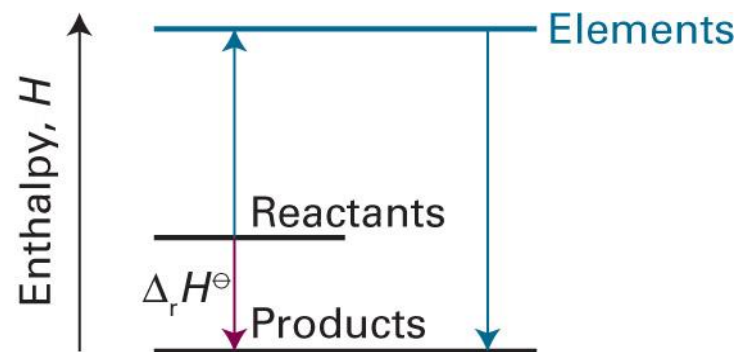
	$\Delta_f H^\ominus / (\text{kJ mol}^{-1})$
$\text{H}_2\text{O}(\text{l})$	-285.83
$\text{H}_2\text{O}(\text{g})$	-241.82
$\text{NH}_3(\text{g})$	-46.11
$\text{N}_2\text{H}_4(\text{l})$	+50.63
$\text{NO}_2(\text{g})$	+33.18
$\text{N}_2\text{O}_4(\text{g})$	+9.16
$\text{NaCl}(\text{s})$	-411.15
$\text{KCl}(\text{s})$	-436.75

	$\Delta_f H^\ominus / (\text{kJ mol}^{-1})$
$\text{CH}_4(\text{g})$	-74.81
$\text{C}_6\text{H}_6(\text{l})$	+49.0
$\text{C}_6\text{H}_{12}(\text{l})$	-156
$\text{CH}_3\text{OH}(\text{l})$	-238.66
$\text{CH}_3\text{CH}_2\text{OH}(\text{l})$	-277.69

(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Tables 2C.4 and 2C.5)



$\Delta_f H^\ominus$ and $\Delta_r H^\ominus$



$$\Delta_r H^\ominus = \sum_{\text{products } i} \nu(i) \Delta_f H^\ominus(i) - \sum_{\text{reactants } j} \nu(j) \Delta_f H^\ominus(j)$$

Temperature Dependence

$$dH = C_p dT$$

$$\int_{\text{state } i}^{\text{state } f} dH = \int_{\text{state } i}^{\text{state } f} C_p dT$$

$$H_f - H_i = \int_{T_i}^{T_f} C_p dT$$

$$H(T_f) - H(T_i) = \int_{T_i}^{T_f} C_p dT$$

Temperature Dependence

For $\nu_A A + \nu_B B \rightarrow \nu_C C + \nu_D D$,

$$\Delta_r H^\ominus(T_f) = \nu_C H_m^\ominus(C, T_f) + \nu_D H_m^\ominus(D, T_f) - \nu_A H_m^\ominus(A, T_f) - \nu_B H_m^\ominus(B, T_f)$$

Since $H(T_f) = H(T_i) + \int_{T_i}^{T_f} C_p dT$,

$$H_m^\ominus(T_f) = H_m^\ominus(T_i) + \int_{T_i}^{T_f} C_{p,m}^\ominus dT$$

$$\Delta_r H^\ominus(T_f)$$

$$= \nu_C \left\{ H_m^\ominus(C, T_i) + \int_{T_i}^{T_f} C_{p,m}^\ominus(C) dT \right\} + \nu_D \left\{ H_m^\ominus(D, T_i) + \int_{T_i}^{T_f} C_{p,m}^\ominus(D) dT \right\}$$

$$- \nu_A \left\{ H_m^\ominus(A, T_i) + \int_{T_i}^{T_f} C_{p,m}^\ominus(A) dT \right\} - \nu_B \left\{ H_m^\ominus(B, T_i) + \int_{T_i}^{T_f} C_{p,m}^\ominus(B) dT \right\}$$

Temperature Dependence

$$\Delta_r H^\ominus(T_f) = \nu_C H_m^\ominus(C, T_i) + \nu_D H_m^\ominus(D, T_i) - \nu_A H_m^\ominus(A, T_i) - \nu_B H_m^\ominus(B, T_i) \\ + \int_{T_i}^{T_f} \underbrace{\{\nu_C C_{p,m}^\ominus(C) + \nu_D C_{p,m}^\ominus(D) - \nu_A C_{p,m}^\ominus(A) - \nu_B C_{p,m}^\ominus(B)\}}_{\Delta_r C_p^\ominus} dT$$

$$\Delta_r H^\ominus(T_f) = \Delta_r H^\ominus(T_i) + \int_{T_i}^{T_f} \Delta_r C_p^\ominus dT$$

Kirchhoff's Law

$$\Delta_r H^\ominus(T_f) = \Delta_r H^\ominus(T_i) + \int_{T_i}^{T_f} \Delta_r C_p^\ominus dT$$

where

$$\Delta_r C_p^\ominus = \sum_{\text{products } i} \nu(i) C_{p,m}^\ominus(i) - \sum_{\text{reactants } j} \nu(j) C_{p,m}^\ominus(j)$$

If $\Delta_r C_p^\ominus$ is constant,

$$\Delta_r H^\ominus(T_f) = \Delta_r H^\ominus(T_i) + \Delta_r C_p^\ominus \times (T_f - T_i)$$

Example 2C.2

The standard enthalpy of formation of $\text{H}_2\text{O}(\text{g})$ at 298 K is $-241.82 \text{ kJ mol}^{-1}$. Estimate its value at 100°C given the following values of the molar heat capacities at constant pressure: $\text{H}_2\text{O}(\text{g})$: $33.58 \text{ J K}^{-1} \text{ mol}^{-1}$; $\text{H}_2(\text{g})$: $28.84 \text{ J K}^{-1} \text{ mol}^{-1}$; $\text{O}_2(\text{g})$: $29.37 \text{ J K}^{-1} \text{ mol}^{-1}$. Assume that the heat capacities are independent of temperature.

The formation reaction is $\text{H}_2(\text{g}) + (1/2)\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{g})$. Hence,

$$\begin{aligned}\Delta_r C_p^\ominus &= C_{p,m}^\ominus(\text{H}_2\text{O}, \text{g}) - C_{p,m}^\ominus(\text{H}_2, \text{g}) - (1/2)C_{p,m}^\ominus(\text{O}_2, \text{g}) \\ &= 33.58 \text{ J K}^{-1} \text{ mol}^{-1} - 28.84 \text{ J K}^{-1} \text{ mol}^{-1} - (1/2) \times (29.37 \text{ J K}^{-1} \text{ mol}^{-1}) \\ &= -9.94 \text{ J K}^{-1} \text{ mol}^{-1}\end{aligned}$$

Example 2C.2

Using Kirchhoff's law,

$$\Delta_r H^\ominus(T_f) = \Delta_r H^\ominus(T_i) + \Delta_r C_p^\ominus(T_f - T_i)$$

$$\begin{aligned}\Delta_r H^\ominus(373 \text{ K}) &= \Delta_r H^\ominus(298 \text{ K}) + \Delta_r C_p^\ominus \times (373 \text{ K} - 298 \text{ K}) \\ &= -241.82 \text{ kJ mol}^{-1} + (-9.94 \text{ J K}^{-1} \text{ mol}^{-1}) \times (75 \text{ K}) \\ &= -242.57 \text{ kJ mol}^{-1}\end{aligned}$$

Summary for Chapter 3 (3)

